

HENRY R. SCHARF

CONTACT INFORMATION

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RESEARCH EMPLOYMENT

Assistant Professor 2023–Current
University of Arizona, Tucson, Arizona

Adjunct Professor 2023–Current
San Diego State University, San Diego, California

Assistant Professor 2019–2023
San Diego State University, San Diego, California

Postdoctoral Fellow 2018–2019
Colorado State University, Fort Collins, Colorado

EDUCATION

Colorado State University, Fort Collins, Colorado USA

PhD in Statistics 2017
Advisor: Mevin Hooten

MS in Statistics 2014

University of Arizona, Tucson, Arizona USA

MEd in Teaching and Teacher Education 2008

BS in Mathematics and Physics 2007
Cum laude with Honors

PUBLICATIONS

Ellwanger, A. L., K. S. Morrow, A. K. Dhawale, **H. R. Scharf**, P. O. Ngakan, E. P. Riley (2026). Balancing risk and reward: Toward resilient human-primate coexistence in a rapidly changing environment in Sulawesi, Indonesia. *Ambio*.

Trinidad, J.*, **H. R. Scharf**, P. O. Ngakan, E. P. Riley (2025). Roadside dining: The collective movement behavior of Sulawesi Moor macaques in a provisioning context. *American Journal of Primatology*, 87: e23727.

Scharf, H. R., J. Schierbaum*, H. Matsumoto, T. Assal (2024). Predicting species-level vegetation cover using large satellite imagery data sets. *Journal of Agricultural, Biological and Environmental Statistics*, 1–20.

McFadden A. J.*, A. D. Stow, P. J. Riggan, R. Tissell, J. O’Leary, **H. R. Scharf** (2024). Estimating Fire Radiative Energy Density with Repeat-Pass Aerial Thermal-Infrared Imaging of Actively Progressing Wildfires. *Fire*, 7(6):179.

Williams, P. J., X. Lu, **H. R. Scharf**, M. B. Hooten (2023). Embracing asymmetry in nature: How to account for skewness in ecological data. *Ecological Informatics*, 75: 102085.

Boulil, Z. L.*, J. W. Durban, H. Fearnbach, T. W. Joyce, S. G. M. Leander, **H. R. Scharf** (2023). Detecting changes in dynamic social networks using multiply-labeled movement data. *Journal of Agricultural, Biological and Environmental Statistics*, 28: 243–259.

Scharf, H. R. (2022). Local indicators of spatial association (LISA). *Wiley StatsRef: Statistics Reference Online*.

Scharf, H. R., X. Lu, P. J. Williams, M. B. Hooten (2022). Constructing flexible, identifiable and interpretable statistical models for binary data. *International Statistical Review*, 90: 328–345.

Raiho, A. M., **H. R. Scharf**, C. A. Roland, D. K. Swanson, S. E. Stehn, and M. B. Hooten (2022). Searching for refuge: A framework for identifying site factors conferring resistance to climate-driven vegetation change. *Diversity and Distributions*, 28(4), 793–809.

Scharf, H. R., A. M. Raiho, S. Pugh, C. A. Roland, D. K. Swanson, S. E. Stehn, M. B. Hooten (2022). Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity. *Biometrics*, 78: 1427–1440.

Reimer, J. R., J. Arroyo-Esquivel, J. Jiang, **H. R. Scharf**, E. M. Wolkovich, K. Zhu, C. Boettiger (2021). Noise can create or erase long transient dynamics. *Theoretical Ecology*, 14: 685–695.

Scharf, H. R. (2021). Statistical analysis of animal movement: Understanding behavior through hierarchical parametric models. *Notices of the American Mathematical Society*, 68(6), 911–924.

Scharf, H. R., F. Buderman (2020). Animal movement models for multiple individuals. *Wiley Interdisciplinary Reviews: Computational Statistics*, e1506.

Scharf, H. R., M. B. Hooten, R. R. Wilson, G. M. Durner, T. C. Atwood (2019). Accounting for phenology in the analysis of animal movement. *Biometrics*, 75: 810–820.

Hooten, M. B., **H. R. Scharf**, J. M. Morales (2019). Running on empty: Recharge dynamics from animal movement data. *Ecology Letters*, 22, 377–389.

Hooten, M. B., **H. R. Scharf**, T. J. Hefley, A. T. Pearse, M. D. Weegman (2018). Animal movement models for migratory individuals and groups. *Methods in Ecology and Evolution*, 9, 1692–1705.

Scharf, H. R., M. B. Hooten, D. S. Johnson, J. W. Durban (2018). Process convolution approaches for modeling interacting trajectories. *Environmetrics*, e2487.

Scharf, H. R., M. B. Hooten, D. S. Johnson (2017). Imputation approaches for animal movement modeling. *Journal of Agricultural, Biological and Environmental Statistics*, 22(3), 335–352.

Hefley, T. J., K. M. Broms, B. M. Brost, F. E. Buderman, S. L. Kay, **H. R. Scharf**, J. R. Tipton, P. J. Williams, and M. B. Hooten. (2017). The basis function approach to modeling autocorrelation in ecological data. *Ecology*, 98(3), 632–646.

Scharf, H. R., M. B. Hooten, B. K. Fosdick, D. S. Johnson, J. M. London, and J. W. Durban.

*Student contributor

(2016). Dynamic social networks based on movement. *Annals of Applied Statistics*, 10(4), 2182–2202.

SPONSORED RESEARCH

<i>Focused Next Steps for Behavioral and Physiological Response Studies with Social Delphinids off Southern California</i> Office of Naval Research, Marine Mammals and Biology. Primary Institution: Southall Environmental Associates, Inc. Sub-Award to UofA: \$130,000. Role: Senior personnel. Status: Terminated 2025-04-03 with \$48,000 remaining.	2024–2026
<i>Factors affecting provisioning and foraging in rapidly changing landscapes</i> National Science Foundation, SBE. Total Award: \$329,849. Role: Co-PI. Status: Funded	2023–2027
<i>Collaborative Research: Pacific Alliance for Low-Income Inclusion in Statistics & Data Science</i> National Science Foundation, S-STEM. Total Award: \$5,000,000. Role: Co-PI 2022–2023. Status: Funded	2022–2028
<i>Visualizing trajectory uncertainty</i> University Grants Program, SDSU. Total Award: \$10,000. Role: PI. Status: Funded	2021–2022
<i>Broadening the impact and accessibility of animal movement models</i> University Grants Program, SDSU. Total Award: \$10,000. Role: PI. Status: Funded	2020–2021

INVITED CONFERENCE PRESENTATIONS

<i>Predicting fine-scale taxonomic variation in landscape vegetation using large satellite imagery data sets</i> EnviBayes Workshop on Complex Environmental Data, Fort Collins, CO, USA	2023
<i>Detecting changes in dynamic social networks using unlabeled movement data</i> Joint Statistical Meetings, Washington D.C., USA	2022
<i>Detecting changes in dynamic social networks based on unlabeled movement data</i> CMStatistics, London, UK	2021
<i>Multivariate Bayesian clustering using covariate-informed components with application to boreal vegetation sensitivity</i> Joint Statistical Meetings, Seattle, WA, USA	2021
<i>Statistical models for dependent trajectories with application to animal movement</i> Joint Statistical Meetings, Denver, CO, USA Topic contributed session for ISBA Savage Award finalists	2019

	<i>Animal movement models for migratory individuals and groups</i> The Wildlife Society Annual Conference, Cleveland, OH Symposium: <i>Animal movement: Advances in movement modeling and their applications</i>	2018
	<i>Imputation approaches for animal movement modeling</i> Joint Statistical Meetings, Vancouver, BC, Canada Invited session	2018
	<i>Dynamic Social Networks Based on Movement</i> Joint Statistical Meetings, Chicago, IL Student paper award winner (ENVR)	2016
DEPARTMENTAL SEMINARS	<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i> Department of Statistics, Pennsylvania State University	2024
	<i>Predicting species-level vegetation cover using large satellite imagery data sets</i> Department of Ecosystems Science and Management, Pennsylvania State University	2024
	<i>Hierarchical statistical models can fuse multiple data sources and account for dependence in observational data</i> School of Natural Resources and the Environment, University of Arizona	2024
	<i>Detecting changes in dynamic social networks based on unlabeled movement data</i> Department of Mathematics and Statistics, California State University Long Beach	2021
	<i>Movement data reveal dynamic social relationships</i> Department of Statistics, University of British Columbia	2021
	<i>Movement data reveal dynamic social relationships</i> Department of Statistics, Kansas State University	2021
	<i>Multi-layered convolutional Gaussian process models for animal movement</i> Computational Sciences Research Center, San Diego State University	2020
	<i>Statistical models for heterogeneous animal movement</i> Department of Statistics & Data Science, University of Texas at Austin	2019
CONTRIBUTED PRESENTATIONS	<i>Wait, who was that? Tracking interacting trajectories with label uncertainty</i> ISBA World Meeting, Nagoya, Japan	2026
	<i>New tools for recursive Bayesian inference applied to high-resolution satellite imagery</i> Joint Statistical Meetings, Nashville, TN, USA	2025

<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i>	2024
Joint Statistical Meetings, Portland, OR, USA	
<i>A strategy for avoiding particle degeneracy in recursive Bayesian inference</i>	2024
Western North American Region of The International Biometric Society annual meeting , Fort Collins, CO, USA	
<i>Identifying species-level vegetation cover using Sentinel-2 imagery</i>	2022
ASA Section on Statistics and the Environment Workshop, Provo, Utah	
<i>Accounting for phenology in the analysis of animal movement</i>	2018
International Statistical Ecology Conference, University of St. Andrews, Scotland	
<i>Process convolution approaches for modeling interacting trajectories</i>	2017
Joint Statistical Meetings, Baltimore, MD	
<i>Process convolution approaches for modeling interacting trajectories</i>	2017
Statistics Department at Colorado State University, Fort Collins, CO. Poster	
<i>Dynamic Social Networks Based on Movement</i>	2016
International Statistical Ecology Conference, Seattle, WA	
<i>Spatiotemporal Models for Animal Social Structure</i>	2015
Joint Statistical Meetings, Seattle, WA	
<i>Spatiotemporal Models for Animal Social Structure</i>	2015
Statistical and Applied Mathematical Sciences Institute ECOL: Transitional Workshop Research Triangle Park, NC	
<i>Dynamic social networks based on movement</i>	2015
Graduate Student Showcase, Fort Collins, CO. Poster	
<i>Spatiotemporal models for animal social structure</i>	2015
Statistics Department at Colorado State University, Fort Collins, CO. Poster	
<i>Novel visualization and analysis for extreme-scale wind turbine simulations</i>	2014
Conference on Data Analysis, Santa Fe, NM. Poster	

TEACHING

<i>Environmental Statistics</i> (Arizona: MATH/STAT/BIOS 574E)	Fall 2024
<i>Design of Experiments</i> (Arizona: MATH/STAT 571B)	Spring 2024
<i>Advanced Statistical Regression Analysis</i> (Arizona: MATH/STAT 571A)	Fall 2023

	<i>R Programming and Data Science</i> (SDSU: STAT 410)	Spring 2023 Fall 2022
	<i>Bayesian Statistics</i> (SDSU: STAT 676)	Spring 2022
	<i>Applied Spatio-Temporal Statistics</i> (SDSU: STAT 596/696)	Spring 2023 Spring 2021 Spring 2020
	<i>Statistical Computing</i> (SDSU: STAT 580)	Fall 2022 Fall 2021 Fall 2020
	<i>Introduction to Statistical Learning</i> (SDSU: STAT 596)	Fall 2019
ADVISING	<i>Masters students</i>	
	Hugo Rosales Portillo Angelica Rivera Navid Nezamabadi Jonathan Schierbaum Zaineb Boulil Jennifer Betancourt Kristine Dinh	Summer 2023 Spring 2023 Summer 2022 Spring 2022 Spring 2021 Spring 2021 Spring 2021
	<i>Undergraduate students</i>	
	Elena Dubocanin	Winter 2022
WORKSHOPS & SHORT COURSES	<i>Workshop: Statistical Models for Animal Movement</i>	
	Co-instructor	
	Day-long workshop on animal movement modeling with examples in R	
	International Association for Landscape Ecology annual meeting Fort Collins, CO, USA	2019
	International Statistical Ecology Conference University of St. Andrews, Scotland	2018
	<i>Training in Bayesian Modeling for Practicing Ecologists.</i>	2016
	Co-instructor	
	NSF-supported two-week workshop Intensive training in Bayesian modeling Colorado State University, Fort Collins, CO	
	<i>R Workshop</i>	2015
	Co-instructor Day-long tutorial on R for ecologists Colorado State University, Fort Collins, CO	
	<i>Tutorial on Parallel Programming in R</i>	2015
	Co-Developer Workshop on Parallel Computing, UC Boulder/CSU	
	<i>Tutorial on Parallel Programming in R</i>	2015
	Co-Developer Conference on Statistical Practice 2015, New Orleans, LA	

AWARDS	ISBA Savage Award - Applied Methodology <i>The Savage Award, named in honor of Leonard J. "Jimmie" Savage, is bestowed each year to two outstanding doctoral dissertations in Bayesian econometrics and statistics.</i> Amount: \$750 + \$500 travel support	2018
	James L., M. Leslie, & Edna Madison Memorial Award <i>This award is given each year to the student selected by the statistics faculty at Colorado State University as the outstanding graduate student in the Department of Statistics and/or the Department of Mathematics</i> Amount: \$675	2017
SERVICE	Award Committee: ISBA Blackwell-Rosenbluth Award for outstanding junior Bayesian researchers	2025
	Scientific Committee: EnviBayes Workshop on Complex Environmental Data	2025
	Program Officer: International Society for Bayesian Analysis Section on Environmental Statistics (EnviBayes)	2023-2024
	Award Committee: ISBA Savage Award - Applied Methodology for outstanding dissertations in Bayesian econometrics and statistics	2021
	Secretary: American Statistical Association Section on Statistics and the Environment (ENVR)	2020
	Treasurer: American Statistical Association Section on Statistics and the Environment (ENVR)	2019
	Associate Editor: <i>Data Science in Science Stat</i>	2021–Current 2023–Current

Reviewer for the following journals:

Annals of Applied Statistics

Advances in Statistical Analytics

Bayesian Analysis

Biometrics

Computational Statistics and Data Analysis

Environmetrics

Environmental and Ecological Statistics

Journal of the American Statistical Association

Journal of Agricultural, Biological, and Environmental Statistics

Journal of Computational and Graphical Statistics

Journal of the Royal Statistical Society: Series C

Network Science

Ecological Monographs

Ecology

Ecology and Evolution

Ecosphere

Methods in Ecology and Evolution

Movement Ecology

Network Science

Proceedings of the Royal Society: Series B

PLOS ONE

SOFTWARE

Scharf, H. R., K. Dinh*, H. Rosales*, A. Rivera* (2025). anipaths: Animation of observed trajectories using spline- or state-space model-based interpolation. R package version 0.10.5.

COMMITMENT TO DIVERSITY

Panelist: 2019 Career Issues Panel Women in Science Symposium

2019

Panel moderator, organizer: Women in Science Symposium

2018

Topics of discussion included:

Practical suggestions for recruiting, retaining, and promoting women

Insight into implicit and explicit bias

Tips to initiate your own Career Issues small group

Networking opportunities with men and women from academic,

industry and the Front Range community to promote gender equality

Career issues

2015–2019

Graduate student-initiated discussion group made up of students, and junior, mid-career, and senior faculty from multiple departments.

Topics included: strategies for career advancement for women,

awareness of systemic sexism in academia, and increasing

participation of women in academia and STEM fields in particular.

OTHER CONFERENCES & WORKSHOPS

Participant in NIMBioS Investigative Workshop *Transients in Biological Systems*.

2019

Member of the working group *Network Models* in the SAMSI 2014-15 Program on Mathematical and Statistical Ecology (ECOL).

2014–2015

*Student contributor

	Pan-American Advanced Studies Institute (Buzios, Brazil) Workshop focused on spatiotemporal statistics, with a special emphasis on international collaboration.	2014
	Design and Analysis of Experiments Conference	2012
PROFESSIONAL AFFILIATIONS	International Biometric Society WNAR region	2017–Current
	International Society for Bayesian Analysis	2017–Current
	Institute of Mathematical Statistics	2015–Current
	American Statistical Association ENVR section	2013–Current